

Federal Funding Agency Requirements for Computational Reproducibility

A Review of NIH and NSF Policies on Data Sharing, Software Documentation, and Reproducible Research

PHB 243b Course Materials

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1 Executive Summary

Federal funding agencies have substantially increased their emphasis on research reproducibility over the past decade. The National Institutes of Health (NIH) and National Science Foundation (NSF) now require explicit plans for data management, sharing, and—to varying degrees—documentation of computational methods. This brief reviews current policies, identifies specific language from agency guidance documents, and discusses implications for biostatisticians and computational researchers.

While neither agency currently mandates specific tools such as Docker or renv, the requirements for transparency, accessibility, and long-term preservation of computational workflows create strong practical incentives for adopting containerization and environment management technologies.

2 NIH Data Management and Sharing Policy

2.1 Policy Overview

The NIH Data Management and Sharing (DMS) Policy (NOT-OD-21-013), effective January 25, 2023, represents the most significant expansion of NIH data sharing requirements in the agency’s history. The policy applies to all research funded or conducted by NIH that generates “scientific data,” defined as:

“Recorded factual material commonly accepted in the scientific community as of sufficient quality to validate and replicate research findings.” (National Institutes of Health 2020a)

The policy explicitly frames data sharing as essential to reproducibility:

“Sharing scientific data accelerates biomedical research discovery, enhances research rigor and reproducibility, provides accessibility to high-value datasets, and promotes data reuse for future research studies.” (National Institutes of Health 2023a)

2.2 Requirements for Software and Code Documentation

The supplemental guidance (NOT-OD-21-014) specifies that Data Management and Sharing Plans must address computational dependencies. Researchers must:

“State whether specialized tools, software, and/or code are needed to access or manipulate shared scientific data, and if so, provide the name(s) of the needed tool(s) and software and specify how they can be accessed.” (National Institutes of Health 2020b)

The guidance further requires researchers to classify tool availability:

- Open source and freely available
- Generally available for purchase in the marketplace
- Available exclusively from the research team

Critically, researchers must address sustainability:

“Plans should note whether such tools are likely to remain available for as long as the scientific data remain available.” (National Institutes of Health 2020b)

This sustainability requirement creates implicit pressure for researchers to use established, well-maintained tools and to document computational environments in reproducible formats.

2.3 Rigor and Reproducibility Requirements

Since 2016, NIH has required grant applications to address four dimensions of rigor and reproducibility (NOT-OD-16-011):

1. **Rigor of prior research:** Assessment of the quality of evidence supporting the proposed work

2. **Rigorous experimental design:** NIH expects “strict application of the scientific method to ensure robust and unbiased experimental design, methodology, analysis, interpretation and reporting of results” (National Institutes of Health 2023b)
3. **Biological variables:** Consideration of sex and other relevant biological variables
4. **Resource authentication:** Verification of key reagents and materials

For computational research, the “rigorous experimental design” criterion implies that methods must be sufficiently documented to allow independent verification.

2.4 Compliance and Monitoring

Beginning October 2024, NIH requires investigators to report compliance with their approved DMS Plans as part of annual progress reports. Non-compliance may result in:

“Special award conditions, enforcement actions, or termination. Post-funding non-compliance affects future funding decisions.” (National Institutes of Health 2023a)

3 NSF Data Management and Sharing Requirements

3.1 General Policy Framework

NSF requires a two-page Data Management and Sharing Plan (DMSP) for all research proposals. The general guidance specifies that plans must address:

“The types of data, samples, physical collections, software, curriculum materials and other materials to be produced in the course of the project.” (National Science Foundation 2024)

NSF explicitly includes software as a research product requiring management and sharing planning.

3.2 Dear Colleague Letter on Reproducibility (NSF 18-053)

In 2018, NSF issued a Dear Colleague Letter emphasizing the agency’s commitment to reproducibility and replicability. The letter encouraged proposals for:

“Research infrastructure for reproducibility and replicability...developing and facilitating adoption of cyberinfrastructure tools and/or research methods that enable use of reproducible and replicable practices across one or more science and engineering communities.” (National Science Foundation 2018)

The letter specifically recommended:

“Use of tools that enable automated capture and recording of experimental and analysis details, such as workflow languages or engines, is encouraged. To allow replication, researchers are encouraged to use interoperable data formats, consider establishing interfaces, and leverage existing open-source analysis environments.” (National Science Foundation 2018)

This language implicitly endorses containerization and environment management approaches, though without mandating specific technologies.

3.3 BIO Directorate Requirements

The NSF Directorate for Biological Sciences (BIO) provides the most explicit guidance on software and code sharing among NSF directorates. BIO’s definition of “data” explicitly includes:

“Software or computer code that is required for replication.” (National Science Foundation, Directorate for Biological Sciences 2024)

Progress reports must document:

“Evidence of research products and their availability, including, but not limited to: data, publications, samples, physical collections, software and models, as described in any Data Management and Sharing Plan.” (National Science Foundation, Directorate for Biological Sciences 2024)

BIO also requires adherence to FAIR principles:

“DMSPs should be consistent with the FAIR (Findability, Accessibility, Interoperability, and Reusability) Guiding Principles for scientific data management and stewardship.” (National Science Foundation, Directorate for Biological Sciences 2024)

3.4 Repository Requirements

NSF explicitly discourages reliance on personal infrastructure:

“Relying on personal or lab websites instead of public repositories may be considered inadequate.” (National Science Foundation, Directorate for Biological Sciences 2024)

Research products require “unique, resolvable, and persistent identifiers (such as digital object identifiers [DOIs]; uniform resource locators [URLs], or similar).”

4 The National Academies Report

4.1 Congressional Mandate

The American Innovation and Competitiveness Act of 2017 directed NSF to commission a National Academies study on reproducibility. The resulting 2019 report, *Reproducibility and Replicability in Science*, established influential definitions and recommendations that have shaped subsequent agency policies.

4.2 Definitions

The report distinguished reproducibility from replicability:

“**Reproducibility** is obtaining consistent results using the same input data; computational steps, methods, and code; and conditions of analysis.” (National Academies of Sciences, Engineering, and Medicine 2019)

“**Replicability** is obtaining consistent results across studies aimed at answering the same scientific question, each of which has obtained its own data.” (National Academies of Sciences, Engineering, and Medicine 2019)

The committee explicitly focused on computational reproducibility:

“The committee’s definition of reproducibility is focused on computation because of its major and increasing role in science...the abundance of data and widespread use of computation have transformed many disciplines. However, this revolution is not yet uniformly reflected in how scientists use software and how scientific results are published and shared.” (National Academies of Sciences, Engineering, and Medicine 2019)

4.3 Key Findings on Computational Reproducibility

The report documented significant failures in computational reproducibility:

“A number of systematic efforts to reproduce computational results across a variety of fields have failed in more than half of the attempts made—mainly due to insufficient detail on digital artifacts such as data, code, and computational workflow.” (National Academies of Sciences, Engineering, and Medicine 2019)

4.4 Recommendations

The report’s recommendations for computational reproducibility include:

“**Recommendation 4-1:** To help ensure the reproducibility of computational results, researchers should convey clear, specific, and complete information about any computational methods and data products that support their published results in order to enable other researchers to repeat the analysis.” (National Academies of Sciences, Engineering, and Medicine 2019)

The report also called for infrastructure investment:

“The NSF should, in harmony with other funders, endorse or create code and data repositories for long-term preservation of digital artifacts. In line with its expressed goal of ‘harnessing the data revolution,’ NSF should consider funding tools, training, and activities to promote computational reproducibility.” (National Academies of Sciences, Engineering, and Medicine 2019)

5 Implications for Containerization and Environment Management

5.1 Gap Between Policy and Practice

Current federal policies emphasize *outcomes* (reproducibility, transparency, long-term accessibility) rather than *methods* (specific tools or technologies). Neither NIH nor NSF mandates Docker, renv, or similar tools. However, the practical requirements for meeting policy goals strongly favor these technologies:

Policy Requirement	Docker/Container Benefit	renv/Environment Manager Benefit
Tool sustainability	Frozen environments persist	Lockfiles capture exact versions
Long-term accessibility	Images archived in registries	Restoration from lockfiles
Platform independence	Containers run anywhere	Package versions specified
Complete documentation	Dockerfile is executable spec	renv.lock is machine-readable

5.2 Emerging Best Practices

The NIH requirement to document whether tools “are likely to remain available for as long as the scientific data remain available” is particularly relevant. Container registries (DockerHub, institutional registries) and CRAN/Bioconductor archives provide more robust long-term preservation than ad hoc documentation.

The NSF BIO directorate’s explicit inclusion of “software or computer code that is required for replication” in the definition of data suggests that code sharing requirements will continue to expand across directorates.

5.3 Training Implications

NIGMS has emphasized that graduate training must include:

“Significant training in quantitative and computational analyses...in-depth training in rigorous experimental design and data interpretation before they embark on their thesis work.” (National Institute of General Medical Sciences 2020)

This creates opportunities for programs to incorporate containerization and environment management into required curricula, preparing students for evolving funder expectations.

6 Conclusions

Federal funding agencies have established clear expectations for data sharing and computational transparency, though specific tool requirements remain undefined. The key policy statements support the adoption of reproducibility tools:

1. **NIH DMS Policy:** Requires documentation of software dependencies and their long-term availability
2. **NIH Rigor and Reproducibility:** Demands sufficient methodological detail for independent verification
3. **NSF DMSP Requirements:** Includes software as a research product requiring sharing plans
4. **NSF BIO Guidance:** Explicitly defines code as “data” for management purposes
5. **National Academies Recommendations:** Calls for infrastructure and training to support computational reproducibility

Researchers who adopt containerization (Docker) and environment management (renv) position themselves to meet these requirements efficiently while future-proofing their work against increasingly stringent expectations.

7 References

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- National Science Foundation, Directorate for Biological Sciences. 2024. “BIO Data Management Plan Guidance.” <https://www.nsf.gov/bio/data-management-plans>.

8 Appendix: Key Policy Documents

8.1 NIH

- NOT-OD-21-013: Final NIH Policy for Data Management and Sharing <https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-013.html>
- NOT-OD-21-014: Supplemental Information to the NIH Policy for Data Management and Sharing: Elements of an NIH Data Management and Sharing Plan <https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-014.html>
- NIH Rigor and Reproducibility Guidance <https://grants.nih.gov/policy-and-compliance/policy-topics/reproducibility/guidance>

8.2 NSF

- NSF Data Management Plan Guidance <https://www.nsf.gov/funding/data-management-plan>

- NSF 18-053: Dear Colleague Letter on Reproducibility and Replicability <https://www.nsf.gov/pubs/2018/nsf18053/nsf18053.jsp>
- NSF BIO Data Management Plan Guidance <https://www.nsf.gov/bio/data-management-plans>

8.3 National Academies

- Reproducibility and Replicability in Science (2019) <https://nap.nationalacademies.org/catalog/25303/reproducibility-and-replicability-in-science>